



SAVE TIME. STAY PREPARED.
WE'VE GOT YOUR BACK.



EMERGENCY SUB PLANS

$$E=mc^2$$

MIDDLE SCHOOL SCIENCE

VOLUME 4

10
NEW
LESSONS

Ten more no-prep emergency lessons for the day you cannot be there. **No lab. No equipment.**



NO-PREP

Just print & go



READY-TO-USE

Complete plans
& materials



FOR BUSY DAYS

Engaging,
easy to follow

- ✓ 10 brand new lessons, none repeated from Volumes 1, 2 or 3
- ✓ No lab, no equipment, no science background needed
- ✓ Every lesson has the words to say and a timing table
- ✓ Runs entirely on paper and a pencil
- ✓ Full answer keys that explain why, not just what

INSIDE THIS VOLUME

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QUALITY
RESOURCES.
CONFIDENT
CLASSROOMS.

Created by a
teacher,
for teachers.



How This Pack Works

START HERE

Ten complete emergency science lessons. Each runs for a full class period, needs nothing but photocopies, and is written so a substitute with no science background can teach it confidently.

This is Volume 4. All ten lessons are new. Nothing here repeats Volumes 1, 2 or 3, so the volumes sit side by side in the drawer and never give a class the same lesson twice.

What every lesson contains

PAGE	WHAT IT IS	WHO IT IS FOR
1	Teacher briefing. Timing table, the exact words to say, the three questions students will ask with the answers written out, and what to do with early finishers.	The substitute. Keep this page.
2	Reading or task page. The material for the lesson, with numbered paragraphs and any diagram needed.	Copy for students.
3	Question page. Six to nine questions with writing space built in.	Copy for students.
4	Go further page. Extension task and a challenge for students who finish early.	Copy for students.
5	Answer key. Every answer explained, plus marking notes and the wrong answers to expect.	The substitute. Keep this page.

ABSOLUTELY NO LAB, AND THAT IS THE POINT

Most schools will not let a substitute run a practical, and rightly so. Equipment and safety supervision need the regular teacher present. That is exactly why science sub plans are hard to find.

Nothing in this pack is heated, poured, cut or plugged in. Every lesson runs on paper. The quick start page says so plainly, and each opening script pre-empts the inevitable question about doing an experiment with a reason that is actually true.

Reading level and grouping

Passages sit in the grades 6 to 8 band. Every lesson works as independent silent work, as partner work, or read aloud by the substitute. The briefing page tells the substitute how to run all three, so no decision is left to guesswork on the day.

HOW THIS PACK RELATES TO THE EARLIER VOLUMES

Volume 1 covers fair tests and variables, cell structure, forces and motion, the periodic table, energy transfers and efficiency, food webs, the particle model and changes of state, the water cycle and weather, plate tectonics, and evaluating data and claims. **Volume 2** covers physical and chemical change, wave properties, heredity and Punnett squares, natural selection, photosynthesis and respiration, the circulatory and respiratory systems, density and buoyancy, series and parallel circuits, moon phases and seasons, and rock layers and relative dating. **Volume 3** covers reflection and refraction, magnets and electromagnets, acids and bases, synthetic materials, the nervous system, digestion, microbes and immunity, evidence for common ancestry, gravity and orbits, and weathering and erosion.

No topic is repeated across the series.

Substitute Quick Start

5 MINUTES • FOR THE SUBSTITUTE

If you have five minutes before the bell, read this page and nothing else. It is enough.

In the first five minutes

- 1 Find the lesson the teacher left, or pick any lesson from the index. They are independent. Order does not matter.
- 2 Copy the three student pages, one set per student.
- 3 Keep the briefing page and the answer key. **Do not photocopy those.**
- 4 Read the briefing page. It has a timing table and the words to say.

DO NOT OPEN THE CUPBOARDS

Nothing in this pack needs equipment, and you should not take anything out of a science storeroom, fridge or cupboard. If a student tells you the class always does a practical on this day, the honest answer is that safety supervision needs their regular teacher present.

How the period runs, every time

MINUTES	WHAT HAPPENS
0-5	Students settle. You read the opening lines from the briefing page out loud. Hand out the student pages.
5-15	Students read the task page. Silent, partner, or read aloud by you. Your choice.
15-40	Students work through the question page. You circulate.
40-50	Anyone finished moves to the Go Further page. Collect everything with two minutes to go.

IF A STUDENT ASKS YOU SOMETHING YOU CANNOT ANSWER

Check the briefing page first. The three most common questions for that lesson are already answered there.

If it is not there, this always works and is always true: **"Write down your best answer and put a star next to it. Your teacher will look at the starred ones."**

Lesson Index

ALL TEN LESSONS IN THIS VOLUME

Lessons are numbered 31 to 40 so they sit alongside the earlier volumes without clashing. Every lesson is independent. Teach them in any order.

#	LESSON	FOCUS	NGSS	TIME
31	Rub, Stick, Spark	Static electricity, charge and attraction	MS-PS2-5	40-50 min
32	Faster, Slower, Stop	Reaction rate, catalysts and energy changes	MS-PS1-6	45-55 min
33	Trading Distance for Force	Levers, ramps, pulleys and mechanical advantage	MS-ETS1-3	45-55 min
34	Ones and Zeros	Digital and analog signals, noise and copying	MS-PS4-3	40-50 min
35	Water Going Up	Plant transport, stomata and transpiration	MS-LS1-3	45-55 min
36	The Instruction Book	DNA, genes, proteins and mutations	MS-LS3-1	45-55 min
37	Sorting the Living World	Classification and dichotomous keys	MS-LS4-2	40-50 min
38	Boom and Crash	Populations, limiting factors and carrying capacity	MS-LS2-1	45-55 min
39	How Far Is Far	Stars, light-years and the scale of the universe	MS-ESS1-3	45-55 min
40	Where the Power Comes From	Natural resources, renewable and non-renewable	MS-ESS3-1	40-50 min

Which lesson should I leave?

- **Planned absence, want real work done:** Lesson 33 or 39. Both produce calculation you can grade.
- **Emergency, class you do not know well:** Lesson 37 or 31. Lesson 37 in particular runs itself and weaker readers do well at it.
- **Restless group, last period:** Lesson 34 or 38. Both have a strong hook and a visible finish line.
- **Out several days:** 32 then 33 run well as a physical science pair, and 37 then 38 do the same for life science.

NGSS codes are given as the closest matching middle school performance expectation. This resource is independently created and is not affiliated with or endorsed by any state agency or standards body.

Rub, Stick, Spark

NO SCIENCE BACKGROUND NEEDED. NO LAB, NO EQUIPMENT.

Everything you need is on this page and the answer key. Nothing in this lesson is heated, poured, cut or plugged in.

Copy for students: the next three pages. **Keep:** this page and the answer key.

What this lesson does

Students work out that rubbing moves electrons rather than making them, then use that to explain why a charged object attracts things that are not charged at all. The lesson kills one very stubborn idea: that something becomes positive by gaining protons.

How the period runs

MIN	WHAT YOU DO
0-5	Read the opening lines below. Hand out all three student pages.
5-13	Students read the page and study the two diagrams.
13-26	Questions 1 to 3. The attract-or-repel table is quick.
26-38	Questions 4 to 6.
38-48	Go Further: the charge ranking list.
48-50	Collect all pages.

SAY THIS TO START

Rub a balloon on your hair and it sticks to the wall. Now think about what that actually means. The wall is not charged. It was not rubbed. Nothing was done to it at all. And yet the balloon holds onto it against gravity. Today you will work out how something charged can pull on something that is not, and along the way you will find out that rubbing does not create any charge whatsoever.

The three questions students will ask

THEY ASK	YOU SAY
"Does rubbing make electricity?"	No. It moves electrons from one surface to the other. Add up the charge on both objects before and after and the total is identical.
"How does something become positive?"	By losing electrons, never by gaining protons. Protons are locked in the nucleus and do not move. This is the one to insist on.
"Why is static worse in winter?"	Dry air. Water in humid air lets charge leak away quietly. In dry air it builds up until it jumps all at once.

FINISHED EARLY?

The Go Further page gives a ranking list of materials. Students use it to predict which of any two rubbed together ends up positive, and which pair would give the biggest charge.

Read: Nothing Is Ever Made

BEFORE YOU READ

Two diagrams. The left one shows what rubbing actually does. The right one explains the trick that makes a charged rod pick up paper. Questions 2 and 5 depend on them.

RUBBING MOVES ELECTRONS. IT DOES NOT MAKE THEM.

BEFORE



Both neutral: equal numbers of each.



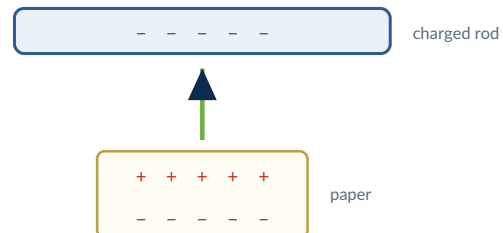
rub them together

AFTER



Two electrons moved across. Six total, before and after.

WHY IT PICKS UP PAPER THAT IS NOT CHARGED



The rod pushes the paper's electrons down to the far side. The near side is left positive, and it is closer, so the pull beats the push. The paper lifts.

- Everything is built from atoms, and atoms contain two kinds of charged particle. **Protons** are positive and sit locked in the nucleus, where they stay. **Electrons** are negative, sit further out, and some of them are held only loosely. An object with equal numbers of each is **neutral**, which is nearly everything nearly all the time.
- Rub two different materials together and some of those loosely held electrons get dragged from one surface onto the other. The object that gained them now has more negative than positive, so it is **negatively charged**. The object that lost them has fewer negatives than positives, so it is **positively charged**.
- Read that again, because it contains the thing most people get wrong. Nothing became positive by gaining anything. It became positive by *losing* electrons. Protons never move between objects. And no charge was created: count up everything before and after and the totals match exactly.
- Charged objects then follow one rule, and it will look familiar. Like charges repel each other. Unlike charges attract. Two balloons rubbed on the same hair will push each other apart, because both ended up with the same charge.
- Now the wall. The wall is neutral, so by that rule nothing should happen. But look at the right-hand diagram. When the negative balloon comes close, it pushes the loosely held electrons in the wall's surface slightly further away. That leaves the near surface of the wall slightly positive and the deeper part slightly negative.
- The balloon is therefore attracted to the near side and repelled by the far side at the same time. But attraction and repulsion both get weaker with distance, and the near side is nearer. The pull wins. That is why a charged object picks up paper, dust, hair and a thin stream of water from a tap, none of which are charged.
- Charge does not stay put forever. It leaks away into the air, and it leaks much faster when the air is humid, because water molecules carry it off. That is the whole reason static shocks are a winter problem: in dry indoor air the charge has nowhere to go, so it accumulates until it finally jumps.
- When it jumps, that is a **spark** — charge forcing its way through air that normally will not conduct at all. Lightning is precisely the same event on an enormous scale, with the charge built up by ice and water churning inside a storm cloud instead of by your shoes on a carpet.

Questions: Rub, Stick, Spark

Answer in the space provided.

1 Explain what actually happens to electrons when you rub a balloon on your hair.

2 A glass rod ends up positively charged. Explain exactly what it gained or lost. Be careful.

Read the third paragraph again before you answer this one.

3 Predict what happens in each case.

WHAT IS BROUGHT TOGETHER	ATTRACT, REPEL OR NOTHING?	WHY
Two negatively charged balloons		
A negative rod and a positive cloth		
Two neutral objects		
A negative rod and a neutral scrap of paper		

4 Charge is never created by rubbing. Explain how you could check that claim by weighing up the charge on both objects before and after.

5 Explain why a charged rod attracts paper even though the paper has no charge of its own.

6 Why do you get more static shocks on a dry winter day than on a humid summer one?

Go Further: Which One Ends Up Positive?

THE RANKING LIST

Materials can be put in order by how tightly they hold their electrons. Rub any two together and the one **higher** in the list loses electrons and becomes positive. The one lower down gains them and becomes negative. The further apart the two materials sit, the bigger the charge.

POSITION	MATERIAL
1 (loses electrons most easily)	Human hair
2	Glass
3	Wool
4	Silk
5	Paper
6	Cotton
7	Hard rubber
8	Polyester
9	PVC plastic
10 (holds electrons most tightly)	Teflon

1 Complete the table using the ranking list.

RUBBED TOGETHER	WHICH BECOMES POSITIVE?	WHICH BECOMES NEGATIVE?
Glass and silk		
Wool and PVC plastic		
Human hair and hard rubber		
Cotton and paper		

2 Which pair in the whole list would produce the **largest** charge if rubbed together? Explain how you know.

3 A balloon rubbed on hair sticks to a wall for about a minute and then falls off. Explain what happened to the charge.

CHALLENGE

A truck carrying fuel drags a metal strap along the road, and the driver must clip a wire from the truck to the ground before pumping any fuel. Fuel moving through a hose builds up charge. Explain what the strap and the wire are for, and why this is a safety rule and not a superstition. Three or four sentences.

Answer Key: Rub, Stick, Spark

QUESTION PAGE

1. Some loosely held electrons are dragged off the hair and onto the balloon. The balloon now has more electrons than protons and is negative; the hair has lost electrons and is positive.
2. It lost electrons. It did not gain protons — protons are locked in the nucleus and never move between objects.

This is the question the lesson exists for. “It gained protons” is the single most common wrong answer in the topic and is worth correcting on the board.
3. Two negative balloons: repel, like charges repel. Negative rod and positive cloth: attract, unlike charges attract. Two neutral objects: nothing. Negative rod and neutral paper: attract, because the rod pushes the paper's electrons away and the near side becomes positive.
4. Measure the charge on both objects together before rubbing and again afterward. The total is the same both times — one object is as negative as the other is positive. Charge only moved.
5. The rod's charge pushes the loosely held electrons in the paper to the far side, leaving the near side oppositely charged. Both attraction and repulsion weaken with distance, and the attracting side is closer, so the pull is stronger than the push and the paper lifts.

Full credit needs the distance argument. A student who says only “it makes the paper charged” has half the answer — the paper is still neutral overall.
6. Humid air carries charge away as fast as it builds up. Dry winter air does not, so charge accumulates until it is large enough to jump as a spark.

60 FURTHER PAGE

RUBBED TOGETHER	POSITIVE	NEGATIVE
Glass and silk	Glass (2)	Silk (4)
Wool and PVC	Wool (3)	PVC (9)
Hair and hard rubber	Hair (1)	Hard rubber (7)
Cotton and paper	Paper (5)	Cotton (6)

2. Human hair and Teflon — positions 1 and 10. They are the furthest apart in the list, and the further apart two materials are, the more electrons transfer and the bigger the charge.
3. The charge leaked away into the air over that minute. Once the balloon was close to neutral again there was nothing to hold it, so it fell.

Challenge. Fuel rubbing through the hose transfers electrons, so charge builds up on the truck and the fuel. If it built up far enough it would eventually jump as a spark, and a spark next to fuel vapor is a fire. The strap and the wire give the charge a continuous path to the ground so it drains away quietly instead of accumulating.

Full credit needs all three steps: charge builds, a spark would jump, the wire drains it harmlessly. This is a real rule at every fuel depot and it is worth saying so.

MARKING SHORTCUT

Mark question 2 and question 5. The first tests whether they understand that only electrons move; the second tests whether they can explain attraction without charge. Between them they cover the whole lesson.

Faster, Slower, Stop

NO SCIENCE BACKGROUND NEEDED. NO LAB, NO EQUIPMENT.

Everything you need is on this page and the answer key. Nothing in this lesson is heated, poured, cut or plugged in.

Copy for students: the next three pages. **Keep:** this page and the answer key.

What this lesson does

Students read two reaction curves off one graph and work out the difference between how fast a reaction goes and how much it produces. They then sort reactions into ones that release heat and ones that absorb it. Keeping speed and amount separate is the whole lesson.

How the period runs

MIN	WHAT YOU DO
0-6	Read the opening lines below. Hand out all three student pages.
6-14	Students read the page and study the graph.
14-28	Questions 1 to 3. Question 1 reads off the graph.
28-40	Questions 4 to 6.
40-53	Go Further: the tablet timings.
53-55	Collect all pages.

SAY THIS TO START

Two identical antacid tablets. One in warm water, one in cold. The warm one fizzes away in about a minute; the cold one takes three. Here is the question that matters, and it is not the obvious one. Which one produced more gas? Almost everyone says the warm one. Almost everyone is wrong, and working out why is today's lesson.

The three questions students will ask

THEY ASK	YOU SAY
"Does a faster reaction make more?"	No. Speed and amount are separate questions. The same tablet makes the same gas whether it takes one minute or ten. It just gets there sooner.
"What does a catalyst actually do?"	It speeds the reaction up and comes out at the end unchanged, so a small amount keeps working indefinitely. It does not get used up.
"Do all reactions give out heat?"	No. Most do, but some absorb it and get genuinely cold. An instant cold pack is a reaction that takes heat in.

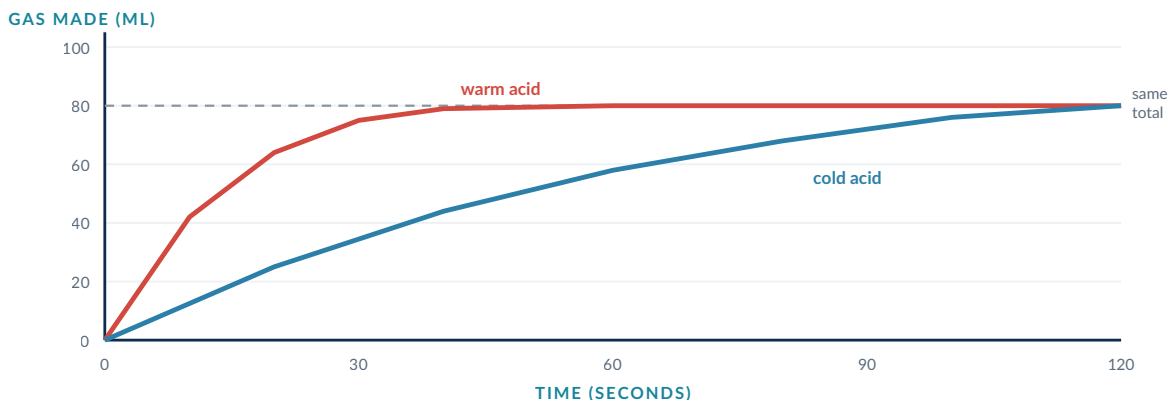
FINISHED EARLY?

The Go Further page has four timings for the same tablet under different conditions. Students identify which change mattered most and predict a fifth.

Read: How Fast, and How Much

BEFORE YOU READ

The same reaction run twice, with only the temperature changed. Question 1 depends entirely on reading this graph, so study it first — particularly where the two lines end up.



1 Reactions have a speed, and the range is enormous. An explosion finishes in a thousandth of a second. Iron rusting through a car door takes a decade. Both are chemical reactions and the only difference is **rate**.

2 Reactions happen when particles crash into each other hard enough to react. Not every collision works — most are too gentle, or the particles hit at the wrong angle and bounce off. So anything that makes collisions more frequent, or harder, makes the reaction faster. That single idea explains every factor below.

3 Temperature. Heat makes particles move faster. They collide more often and they collide harder, so more of those collisions are successful. A rough rule is that a ten degree rise roughly doubles the rate, which is why a refrigerator works: it does not stop food spoiling, it just slows the reactions down.

4 Concentration. More particles crowded into the same space means more collisions per second. Stronger acid reacts faster than weak acid for exactly that reason.

5 Surface area. Only particles at a surface can be hit. A solid lump has almost all of its particles buried inside where nothing can reach them. Crush it to powder and you expose all of them at once. This is why flour, custard powder and sawdust can explode when a lot of it is suspended in air, and why the same material in a sack simply sits there.

6 A catalyst is a substance that speeds a reaction up and is left unchanged at the end. It is not a fuel and it is not used up, so a tiny amount goes on working indefinitely. Your body runs on catalysts — every enzyme is one — and so does the catalytic converter in a car exhaust.

7 Separately from all of that, reactions do something with energy. Most release it into the surroundings and warm them up: burning, rusting, an explosion, a hand warmer. These are **exothermic**. Some absorb energy and the surroundings get colder. These are **endothermic**, and an instant cold pack is one you can buy.

8 Keep those two ideas apart, because they are independent. Rusting is exothermic and takes years. An instant cold pack is endothermic and works in seconds. Fast does not mean hot, and slow does not mean cold. And crucially, none of the four factors above changes how much product you end up with. Look again at where the two lines on the graph finish.

Questions: Faster, Slower, Stop

Questions 1 and 2 refer to the graph on the reading page.

- 1 Which reaction was faster, and how can you tell from the graph? Quote one pair of numbers.

- 2 Both lines finish at the same height. Explain what that tells you, and why it surprises most people.

This is the most important question on the page.

- 3 Complete the table. For each factor, explain the effect using the idea of collisions.

FACTOR	EFFECT ON RATE	WHY, IN TERMS OF COLLISIONS
Raising the temperature		
Increasing concentration		
Crushing a lump into powder		
Adding a catalyst		

- 4 A sack of flour sits safely in a warehouse, but flour dust suspended in the air of a mill can explode. Explain the difference.

- 5 Sort these: burning wood, an instant cold pack, rusting iron, a hand warmer, photosynthesis. Write **EXO** or **ENDO** beside each.

- 6 Rusting releases heat but takes years. Explain how a reaction can be exothermic and slow at the same time.

Go Further: Four Tablets, Four Times

EXTEND

THE DATA

The same brand of effervescent tablet, dropped into the same volume of water, timed until no fizz remains. Everything except the two named conditions was kept identical.

TEST	TABLET	WATER TEMPERATURE	TIME TO FINISH
A	Whole	40°F	180 s
B	Whole	100°F	65 s
C	Crushed	40°F	70 s
D	Crushed	100°F	25 s

- 1 Compare A with B, then A with C. Which single change made the bigger difference on its own?
- 2 Test D combines both changes. Is its time roughly what you would expect from the other three? Show your reasoning.
- 3 Predict roughly how long a crushed tablet would take in ice water at 32°F, and explain your reasoning.
- 4 All four tests used one identical tablet. What was the same in every test besides the tablet, and why does that matter?

CHALLENGE

The packet claims the tablet “releases the same amount of gas however you take it.” Test D finished seven times faster than test A. Explain why the claim can still be true, and say what the graph on the reading page would look like for all four tests drawn together. Three or four sentences.

Answer Key: Faster, Slower, Stop

QUESTION PAGE

1. The warm acid. Its line rises far more steeply — it had made about 64 mL by 20 seconds while the cold acid had made only about 25 mL.
2. It means both reactions produced exactly the same total amount of gas. Temperature changed how quickly it arrived, not how much there was. People expect the faster one to make more because they run speed and amount together, but the amount is fixed by how much material there was to react.
This is the conceptual heart of the lesson. Full credit needs the idea that the amount is set by the ingredients, not by the conditions.
3. Temperature: faster, because particles move faster so they collide more often and harder. Concentration: faster, because there are more particles in the same space so more collisions per second. Powder: faster, because far more particles are exposed at a surface where they can be hit. Catalyst: faster, and it is not used up.
4. In a sack, nearly all the flour particles are buried inside the pile with almost no surface exposed. Suspended in air, every particle has its whole surface available at once, so the reaction can happen everywhere simultaneously.
5. EXO: burning wood, rusting iron, hand warmer. ENDO: instant cold pack, photosynthesis.
Rusting catches people out because it never feels warm. The heat is real; it is just released so slowly that it disperses unnoticed.
6. Exothermic describes what the reaction does with energy; slow describes how quickly it happens. They are separate. Rusting releases heat, but so gradually that it spreads into the surroundings without ever being noticeable.

GO FURTHER PAGE

1. A to B (heating) took 180 s down to 65 s, a saving of 115 s. A to C (crushing) took 180 s down to 70 s, a saving of 110 s. Heating was very slightly more effective, but the two are close to equal.
Accept either as the answer provided the numbers are compared. The point is the comparison, not the winner.
2. Yes. Each change on its own cut the time to roughly a third. Doing both should cut it to roughly a third of a third, or about a ninth, and $180 \div 9 = 20$ s. The measured 25 s is close to that.
Strong students notice the two effects multiply rather than add. Adding the savings would predict a negative time, which is a useful thing to point out.
3. Longer than 70 s. Crushed in 40°F water took 70 s, and 32°F is colder still, so the particles move more slowly and collide less often. Somewhere around 85 to 100 s is a sensible estimate.
Accept any answer above 70 s with reasoning. The reasoning is the mark.
4. Same brand and size of tablet, same volume of water, same container. It matters because if anything else had changed you could not tell which change caused the difference in time.

Challenge. The claim is about amount, not speed, and none of the four changes alters how much material is in the tablet. Drawn together, all four lines would rise at different steepnesses but flatten out at exactly the same height. D would climb steeply and level off early; A would climb gently and reach the same level much later.

This ties the graph to the table and is the best question in the lesson. Full credit needs “same final height, different steepness”.

Trading Distance for Force

NO SCIENCE BACKGROUND NEEDED. NO LAB, NO EQUIPMENT.

Everything you need is on this page and the answer key. Nothing in this lesson is heated, poured, cut or plugged in.

Copy for students: the next three pages. **Keep:** this page and the answer key.

What this lesson does

Students calculate mechanical advantage for four levers, then discover from a data table that force multiplied by distance comes out the same for every method of moving the same load. Nothing is gained; it is only ever redistributed, and the data makes that unavoidable.

How the period runs

MIN	WHAT YOU DO
0-6	Read the opening lines below. Hand out all three student pages. Work through the lever diagram out loud.
6-14	Students read the page.
14-28	Questions 1 to 3. The calculation table is the main task.
28-40	Questions 4 to 6.
40-53	Go Further: five ways to move a piano.
53-55	Collect all pages.

SAY THIS TO START

A machine lets you lift something you could not possibly lift. One person with a long enough bar can raise the corner of a car. So where does the extra strength come from? It does not come from anywhere, and that is the point. You are not getting more for nothing. You are paying for it in a currency you probably have not noticed, and today you will find out what it is.

The three questions students will ask

THEY ASK	YOU SAY
"Does a machine reduce the work?"	No. It changes how the work is spread out. Less force over more distance, or more force over less. Multiply them and you get the same answer.
"What is mechanical advantage?"	How many times the machine multiplies your force. Load divided by effort. An advantage of 4 means you push with a quarter of the weight.
"Why not just make the lever longer and longer?"	You can, and the force keeps dropping. But your end has to travel further every time, and eventually the bar bends or you run out of room.

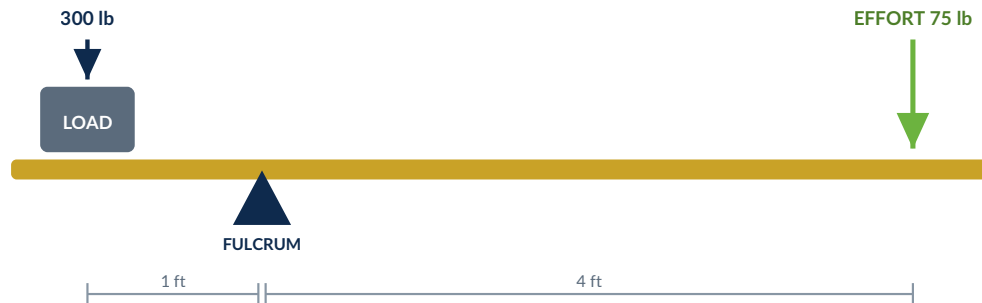
FINISHED EARLY?

The Go Further page gives five different ways of getting the same piano onto the same stage, each with its force and distance. Students multiply, and the answer is identical every time.

Read: Nothing Is Free

BEFORE YOU READ

One lever, drawn with its measurements. Every calculation on the next page uses the same two numbers this diagram shows: how far the load sits from the fulcrum, and how far you are.



Four times the distance from the fulcrum, so one quarter of the force — but your end travels four times as far.

- 1 A machine does not make you stronger. What it does is let you apply a smaller force over a longer distance instead of a large force over a short one. That trade is the entire subject, and every machine in this lesson is a different way of making it.
- 2 Start with the **lever**: a rigid bar that turns on a fixed point called the **fulcrum**. The load sits on one side and you push on the other. What decides how easy it is has nothing to do with the length of the bar itself — it is the ratio of the two distances from the fulcrum.
- 3 In the diagram the load is 1 foot from the fulcrum and you are 4 feet from it. That is four times as far, so you need one quarter of the force: 75 pounds instead of 300. The number of times a machine multiplies your force is its **mechanical advantage**, and here it is 4.
- 4 You can calculate it two ways and both give the same answer. Divide the load by your effort ($300 \div 75 = 4$). Or divide your distance from the fulcrum by the load's distance ($4 \div 1 = 4$). The second is more useful, because you can work out the force you will need before you try.
- 5 Here is the cost. To raise the load by 1 inch, your end of the bar has to travel 4 inches. A quarter of the force, four times the distance. Multiply force by distance on either side and you get exactly the same number. Nothing was created; it was rearranged.
- 6 A **ramp** does the same trade without any moving parts. Pushing a barrel straight up 3 feet takes its full weight. Push it up a 12 foot ramp to the same height and you need a quarter of the force, because you travel four times as far. A staircase is a ramp cut into steps, and a mountain road switching back and forth is a very long ramp.
- 7 A **pulley** system splits the load between the ropes actually supporting it. Two supporting ropes, half the force. Four, a quarter. And once again you pay: with four ropes you must pull four feet of rope through your hands to raise the load one foot.
- 8 One honest correction. Real machines always need slightly more effort than the arithmetic says, because some of what you put in goes into friction and into flexing the machine itself rather than into moving the load. The trade is real, but the exchange rate is never quite perfect.

Questions: Trading Distance for Force

Answer in the space provided. Show your working for any calculation.

1 Complete the table. Mechanical advantage = effort distance ÷ load distance.

LEVER	LOAD DISTANCE	EFFORT DISTANCE	MECHANICAL ADVANTAGE	EFFORT NEEDED TO LIFT 400 LB
A	1 ft	2 ft		
B	1 ft	5 ft		
C	2 ft	8 ft		
D	3 ft	3 ft		

2 Lever D has a mechanical advantage of 1, so it saves no force at all. Suggest why anyone would ever use one.

3 Explain, in your own words, what you pay in exchange for needing less force.

4 A barrel is pushed up a ramp instead of lifted straight up. Explain why a longer ramp needs less force, using the idea of the trade.

5 A pulley system has 4 ropes supporting the load. How much force is needed to lift 200 lb, and how much rope must be pulled to raise it 2 feet?

6 In a real workshop, lifting 400 lb with a mechanical advantage of 4 needs about 115 lb rather than the 100 lb the arithmetic predicts. Explain where the extra 15 lb goes.

Go Further: Five Ways to Move a Piano

THE SITUATION

A 400 pound piano has to be raised 3 feet onto a stage. Five different methods are available. For each one, the force needed and the distance you must move have been measured.

METHOD	FORCE NEEDED	DISTANCE YOU MOVE	FORCE × DISTANCE
Lift it straight up	400 lb	3 ft	
Short ramp	200 lb	6 ft	
Long ramp	100 lb	12 ft	
Pulley, 2 supporting ropes	200 lb	6 ft	
Pulley, 4 supporting ropes	100 lb	12 ft	

- 1 Complete the last column. Write down what you notice.
- 2 Which method needs the least force? Which involves the least moving? Can any method give you both?
- 3 Explain what your answer to question 1 shows about what a machine actually does.
- 4 A sixth method is proposed: a very long ramp needing only 50 lb. How far would you have to push the piano? Show your working.

CHALLENGE

In practice the long ramp needs about 115 lb, not the 100 lb in the table, and the four-rope pulley needs about 125 lb. Both should have been 100. Explain where the extra force is going, and suggest why the pulley loses more of it than the ramp does. Three or four sentences.

Answer Key: Trading Distance for Force

QUESTION PAGE

LEVER	MECHANICAL ADVANTAGE	EFFORT FOR 400 LB
A	$2 \div 1 = 2$	200 lb
B	$5 \div 1 = 5$	80 lb
C	$8 \div 2 = 4$	100 lb
D	$3 \div 3 = 1$	400 lb

2. Because a lever can be useful for something other than saving force — it can change the direction of a push, or move the load further and faster than your hand moves. A pair of scissors and a fishing rod both work this way.

Accept any sensible answer. Students who say "it is useless" have applied the arithmetic correctly and stopped thinking — worth pushing.

3. Distance. Whatever fraction of the force you need, you have to move that many times further.

4. A longer ramp covers the same height over a greater distance, so the same total work is spread over more feet and each foot needs less force.

5. 50 lb, because the load is shared between 4 ropes. You must pull 8 feet of rope to raise it 2 feet.

The second half catches people. Four ropes means four feet of rope for every foot of lift.

6. Into friction and into bending or stretching the machine itself. That effort still has to be supplied but it does not end up lifting the load.

60 FURTHER PAGE

METHOD	FORCE × DISTANCE
Lift straight up	$400 \times 3 = 1,200$
Short ramp	$200 \times 6 = 1,200$
Long ramp	$100 \times 12 = 1,200$
Pulley, 2 ropes	$200 \times 6 = 1,200$
Pulley, 4 ropes	$100 \times 12 = 1,200$

1. Every method gives 1,200. The number does not change no matter which method is used.

2. Least force: the long ramp or the four-rope pulley, both 100 lb. Least moving: lifting it straight up, only 3 feet. No method gives both, because reducing one always increases the other.

3. That a machine does not reduce the total amount that has to be done. It only lets you choose how to pay — a big force over a short distance, or a small force over a long one. The product is fixed at 1,200 whatever you choose.

This is the question the whole page exists for. Full credit needs the idea that the total is fixed and only its distribution changes.

4. $1,200 \div 50 = 24$ feet.

Challenge. The extra force goes into friction and into flexing the equipment, neither of which lifts the piano. The pulley loses more because it has many moving parts — several wheels turning on axles and a rope bending round each one, each adding friction. A ramp has a single sliding surface.

Accept any answer identifying friction plus a reason the pulley has more of it. Strong students mention the rope having to bend, which is a real and often overlooked loss.

Ones and Zeros

NO SCIENCE BACKGROUND NEEDED. NO LAB, NO EQUIPMENT.

Everything you need is on this page and the answer key. Nothing in this lesson is heated, poured, cut or plugged in.

Copy for students: the next three pages. **Keep:** this page and the answer key.

What this lesson does

Students compare a wave that carries information by matching it with a signal that carries information as two states, then work out why only one of them can be copied perfectly. It explains something every student has noticed – that digital video is either perfect or gone, never grainy.

How the period runs

MIN	WHAT YOU DO
0-5	Read the opening lines below. Hand out all three student pages.
5-13	Students read the page and study the four signals.
13-26	Questions 1 to 4.
26-38	Questions 5 and 6.
38-48	Go Further: the copy test.
48-50	Collect all pages.

SAY THIS TO START

Copy a photocopy, then copy the copy, and keep going. By about the tenth generation it is unreadable. Now copy a file on a computer ten thousand times. The last one is identical to the first, byte for byte, with nothing lost at all. Same idea, copying something, and completely different outcomes. Today you find out what digital actually means, and why it has that property.

The three questions students will ask

THEY ASK	YOU SAY
“What does analog mean?”	The signal is a continuous copy of the original – a wiggle shaped like the sound wave that made it. Digital means the signal is only ever one of two values.
“Why can't you just filter the noise out?”	Because in an analog signal the noise looks exactly like signal. There is no way to tell which wiggles were the music and which were interference.
“Is digital always better?”	For copying and sending, yes. The cost is that you have to chop the original into steps, so you never store the true continuous original.

FINISHED EARLY?

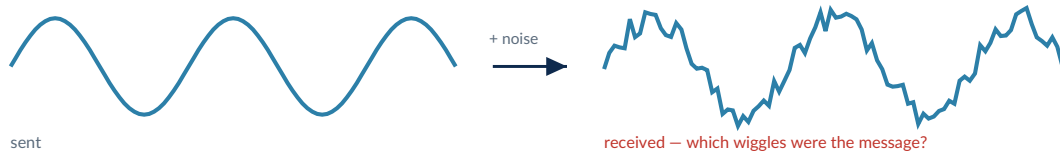
The Go Further page has quality scores for repeated copying of an analog and a digital recording. The pattern is stark and the challenge question explains digital television.

Read: Two Ways to Carry a Message

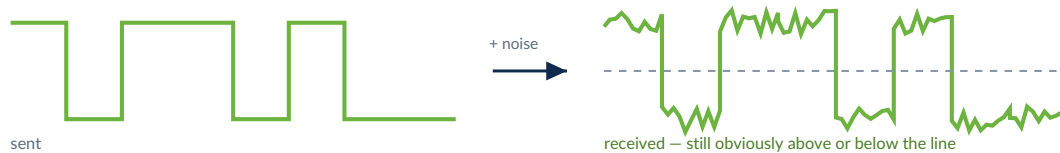
BEFORE YOU READ

Four signals. On each row, the same signal is shown as it was sent and as it arrived after picking up interference. Questions 2 and 3 depend on comparing the two rows.

ANALOG – A COPY OF THE ORIGINAL WIGGLE



DIGITAL – ONLY EVER HIGH OR LOW



- 1 Information always has to be carried by something physical. A sound is carried by air pressure changing; a phone call is carried by a voltage changing in a wire or a radio wave changing in the air. The question this lesson asks is what shape you make those changes, and it turns out there are only two useful answers.
- 2 An **analog** signal is a continuous copy. The voltage in the wire rises and falls in exactly the same shape as the sound wave that produced it. A vinyl record does it physically: the groove wiggles side to side in the same shape as the music. It is an elegant idea and it works well.
- 3 It has one fatal weakness. Every wire, every mile of cable and every amplifier adds a small amount of random interference, called **noise**. And noise in an analog signal is a wiggle. The message is also a wiggle. Once they are added together there is no way to tell them apart, because there is nothing to compare against.
- 4 So the damage accumulates. Copy an analog recording and you copy the noise and add fresh noise on top. Copy that and you do it again. After enough generations the noise is louder than the music, which is exactly why the tenth photocopy is unreadable.
- 5 A **digital** signal takes a completely different approach. The signal is only ever allowed to be one of two values: high or low, on or off, 1 or 0. Nothing in between is permitted or meaningful. The message is carried by the pattern of those two states rather than by the shape of a wave.
- 6 Noise still gets added – look at the bottom right of the diagram. But now it does not matter, because the receiver is not trying to reproduce the shape. It only has to answer one question for each moment: was this above the line or below it? As long as the answer is still obvious, the original pattern can be recovered exactly and a clean new signal sent onward.
- 7 That is the whole trick, and it is worth stating plainly: the receiver does not repair the damaged signal. It reads what was meant and builds a brand new perfect one. Which is why a file copied ten thousand times is identical to the original, and why digital television does not go grainy.
- 8 Nothing is free. To go digital you must chop the smooth original into a series of steps and record a number for each one, and no number of steps is ever quite the true continuous original. Use enough of them – a CD takes 44,100 measurements every second – and no human ear can detect the difference.

Questions: Ones and Zeros

Questions 2 and 3 refer to the diagram on the reading page.

1 Explain the difference between an analog signal and a digital signal in your own words.

2 Look at the top right of the diagram. Explain why the noise cannot be removed from that signal.

3 Look at the bottom right. The digital signal has just as much noise on it. Explain why it does not matter.

This is the most important question on the page.

4 Classify each one. Write **A** for analog or **D** for digital.

EXAMPLE	A OR D	EXAMPLE	A OR D
A vinyl record groove		A text message	
A cassette tape		A CD	
A traffic light		A mercury thermometer	

5 Explain why a photocopy of a photocopy gets worse, but a copy of a computer file does not.

6 Going digital has a cost. Explain what is lost when a smooth original is turned into digital, and why it usually does not matter.

Go Further: The Copy Test

THE DATA

The same three minutes of music was recorded twice, once on tape and once as a computer file. Each was then copied repeatedly, always from the previous copy rather than from the original. Quality was scored out of 100, where 100 means identical to the original.

NUMBER OF COPIES MADE	1	5	10	20	50
Tape (analog)	96	78	61	38	9
File (digital)	100	100	100	100	100

- 1 Describe what happens to the tape recording as copies are made. Use numbers.
- 2 Between which two counts did the tape first drop below half its original quality?
- 3 Explain why the digital row does not change at all, even after fifty copies.
- 4 Suggest why the tape scored 96 rather than 100 even on the very first copy.

CHALLENGE

An analog television with a weak signal gives a grainy, snowy picture that gets gradually worse as the signal weakens. A digital television with a weak signal is perfect, then freezes into blocks, then shows nothing at all – there is no gradual middle. Explain both behaviors using what you have learned. Three or four sentences.

Answer Key: Ones and Zeros

QUESTION PAGE

1. An analog signal is a continuous copy that changes in the same shape as the original. A digital signal is only ever one of two values, high or low, and carries the message as a pattern of those two.
2. Because the noise is a wiggle and the message is a wiggle. There is nothing to distinguish them and no original to compare against, so removing the noise would mean removing part of the message.
3. Because the receiver does not have to reproduce the shape. It only has to decide, for each moment, whether the signal was above or below the line. The noise is far too small to change that answer, so the original pattern is recovered exactly and a clean new signal is sent on.

Full credit needs the idea that a decision is being made rather than a repair. Students who say “the noise is filtered out” have the wrong mechanism, even though the outcome is right.

4. Analog: vinyl record, cassette tape, mercury thermometer. Digital: text message, CD, traffic light.

The traffic light and the thermometer are the interesting pair. A traffic light has a fixed number of states; a mercury column can sit at any height at all.

5. A photocopy is analog, so each generation copies the previous noise and adds more, and the damage builds up. A computer file is digital, so each copy is read as a pattern of ones and zeros and a fresh perfect version is written. No damage accumulates because nothing is being traced.
6. The smooth original has to be chopped into steps and a number recorded for each, so the stored version is a very close approximation rather than the true continuous signal. With enough steps — 44,100 a second on a CD — the gaps are far too small for a human to detect.

60 FURTHER PAGE

1. It gets steadily worse with every copy: 96 after one, 61 after ten, and only 9 after fifty. By the fiftieth copy almost nothing of the original is left.
2. Between 10 copies (61) and 20 copies (38). It crossed 50 somewhere in between.
3. Because each copy is made by reading the pattern of ones and zeros and writing a brand new perfect copy, rather than tracing the previous one. Nothing accumulates, so the fiftieth copy is produced from exactly the same pattern as the first.
4. Because even a single copy adds some noise. The tape machine, the wire and the tape itself all add a small amount, and once added it cannot be separated out again.

Challenge. Analog television sends the picture as a continuous signal, so noise is added directly to the picture and shows up as snow. Weaker signal means more noise relative to picture, so it degrades gradually. Digital television only has to decide between high and low. While the signal is strong enough for that decision the picture is perfect, but once noise makes the decision unreliable the receiver cannot guess — it either shows the last good block or nothing. There is no gradual middle because there is no gradual signal.

This is the payoff of the lesson and explains something every student has seen. Full credit needs the cliff-edge idea: the decision either works or it does not.

IF A STUDENT FINISHES EVERYTHING

Ask them why digital systems often send a few extra bits alongside the message so the receiver can check whether what arrived makes sense. What could a receiver do with that information that an analog receiver could never do?

Water Going Up

NO SCIENCE BACKGROUND NEEDED. NO LAB, NO EQUIPMENT.

Everything you need is on this page and the answer key. Nothing in this lesson is heated, poured, cut or plugged in.

Copy for students: the next three pages. **Keep:** this page and the answer key.

What this lesson does

Students work out how a tree lifts water three hundred feet with no pump, then meet the trade-off that governs a plant's whole life: the same holes that let carbon dioxide in let water out. The Go Further data makes that trade-off numerical, which is where it stops being a slogan.

How the period runs

MIN	WHAT YOU DO
0-6	Read the opening lines below. Hand out all three student pages.
6-14	Students read the page.
14-28	Questions 1 to 3. The xylem and phloem table is the core.
28-40	Questions 4 to 6.
40-53	Go Further: the stomata trade-off data.
53-55	Collect all pages.

SAY THIS TO START

A redwood lifts water more than three hundred feet into the air, every day, all day. It has no pump, no heart, no muscles and no moving parts of any kind. There is no motor anywhere in a tree. So what is pulling the water up? The answer is genuinely strange, and it is happening in every houseplant in this building right now.

The three questions students will ask

THEY ASK	YOU SAY
"Do roots push the water up?"	Barely. Roots do push a little, but nothing like enough for a tall tree. Almost all of the lift comes from the top, not the bottom.
"What is transpiration?"	Water evaporating out of the leaves. Because water molecules stick to each other, water leaving the top drags the whole column up behind it.
"Why doesn't the plant just close the holes?"	Because the same holes let carbon dioxide in, and without that it cannot make food. Closing them saves water and stops it eating.

FINISHED EARLY?

The Go Further page gives water lost and sugar made at five different stomatal openings. Students find the most water-efficient setting and discover it is not the one that makes the most food.

Read: The Engine Is at the Top

- 1 Every cell in a plant needs water, and the ones that need it most are the leaves, which are as far from the ground as it is possible for the plant to put them. In a tall tree that is a lift of hundreds of feet, achieved continuously, with no pump of any kind. Understanding how is worth more than memorizing the parts.
- 2 Water enters through the roots, and the surface area problem is solved the same way it is in your intestine: with enormous numbers of tiny projections. Root hairs are single cells extending out into the soil, and a single rye plant has been measured with hundreds of miles of them.
- 3 From there it travels in **xylem**, which are tubes running the whole height of the plant. Xylem cells are dead and hollow with their end walls dissolved away, which makes them essentially drinking straws made of reinforced tube. They carry water and dissolved minerals, upward only, and never anything else. Wood is mostly old xylem.
- 4 Alongside them run **phloem** tubes, and these are different in every respect. Phloem cells are alive. They carry dissolved sugar made in the leaves, and they carry it in whichever direction it is needed — down to the roots in summer, up to new buds in spring. Xylem is one-way plumbing; phloem is a delivery network.
- 5 Now the lift. Leaves are covered in microscopic holes called **stomata**. Water evaporates out of them constantly, and here is the crucial fact: water molecules stick strongly to one another. So when a molecule leaves the top of the column, it pulls the next one up behind it, and that one pulls the next, all the way down to the root. The whole column is dragged upward like a rope being hauled from above.
- 6 That is **transpiration pull**, and it means the engine is at the top, not the bottom. The tree is not pushing water up. The atmosphere is pulling it, and the tree simply provides unbroken tubes for it to be pulled through.
- 7 Which sets up the trade-off that governs the entire life of a plant. Stomata have to be open to let carbon dioxide in, and without carbon dioxide there is no food. But every second they are open, water escapes. A plant cannot have one without the other, and around ninety-five percent of all the water a plant takes up is lost this way.
- 8 So plants manage the compromise minute by minute. Guard cells on either side of each stoma swell to open it and go limp to close it. In fierce heat a plant will close them to survive, accepting that it stops making food while they are shut. And when there is not enough water to keep the cells firm, the whole structure goes soft and the plant wilts.

Questions: Water Going Up

Answer in the space provided.

1 Complete the comparison table.

	XYLEM	PHLOEM
What it carries		
Which direction		
Alive or dead?		

2 Explain how water gets to the top of a tall tree when the plant has no pump.

This is the most important question on the page.

3 Root hairs are tiny and there are enormous numbers of them. Explain why that matters.

4 Explain the trade-off a plant faces every time it opens its stomata.

5 On a very hot day a plant closes its stomata. State one benefit and one cost of doing that.

6 Explain what is actually happening inside a plant that has wilted.

Go Further: The Trade Every Plant Makes

THE DATA

The same plant measured at five different stomatal openings over one hour, under identical light and temperature. Units are arbitrary — only the comparison matters.

STOMATA OPEN	WATER LOST PER HOUR	SUGAR MADE PER HOUR	SUGAR PER UNIT OF WATER
0% (closed)	2	0	
25%	30	22	
50%	70	38	
75%	120	46	
100%	180	48	

- 1

Describe what happens to water loss and to sugar production as the stomata open wider.
- 2

Complete the last column by dividing sugar by water. Which opening is the most efficient use of water?
- 3

Going from 75% to 100% open costs 60 more units of water. What does the plant gain for it? Was it worth it?
- 4

A plant is growing where water is plentiful and the growing season is short. Which opening would you expect it to use, and why?

CHALLENGE

A cactus keeps its stomata closed all day and opens them only at night, storing what it takes in until morning. Night air is cool and damp; day air is hot and dry. Explain the advantage this gives the cactus, and explain the cost — why every plant does not do it. Three or four sentences.

Answer Key: Water Going Up

QUESTION PAGE

1. Xylem: water and dissolved minerals; upward only; dead hollow cells. Phloem: dissolved sugar; both directions, wherever it is needed; living cells.
2. Water evaporates out of the stomata in the leaves. Water molecules stick strongly to each other, so a molecule leaving the top pulls the next one up behind it, and that pull is transmitted all the way down an unbroken column of water in the xylem to the roots. The pull comes from the top, not a push from the bottom.
Full credit needs both the evaporation and the fact that water molecules stick together. Without the second, there is nothing to transmit the pull.
3. They give an enormous total surface area in contact with the soil, so far more water and mineral can be absorbed than a smooth root could manage.
4. Stomata must be open to let carbon dioxide in, and without carbon dioxide the plant cannot make food. But whenever they are open, water evaporates out. The plant cannot get one without paying the other.
5. Benefit: it stops losing water, which may be what keeps it alive. Cost: no carbon dioxide can get in, so it cannot make food while they are shut.
6. There is not enough water in the cells to keep them firm and pressed against each other, so they go soft and the plant loses the internal pressure that was holding it upright.

60 FURTHER PAGE

OPEN	WATER	SUGAR	SUGAR PER UNIT WATER
0%	2	0	0
25%	30	22	0.73
50%	70	38	0.54
75%	120	46	0.38
100%	180	48	0.27

1. Both increase, but not in the same way. Water loss keeps climbing steeply all the way to 180. Sugar rises quickly at first and then almost levels off, gaining only 2 units between 75% and 100%.
2. 25% open is the most efficient, at 0.73 sugar per unit of water. Efficiency falls steadily as the stomata open further.
Worth pointing out: the most efficient setting is also the one that makes the least food. Efficient and best are not the same thing, and which one matters depends on how much water there is.
3. Only 2 extra units of sugar for 60 extra units of water. It is a very poor trade — unless water is unlimited and every scrap of sugar counts.
4. Fully open, or close to it. With plenty of water there is little penalty for losing it, and a short season means making the most sugar quickly matters more than making it efficiently.

Challenge. Opening at night means the air is cool and damp, so far less water evaporates for the same amount of carbon dioxide taken in. That is a huge saving in a desert. The cost is that photosynthesis needs light, so the cactus has to store the carbon dioxide overnight and use it the next day, which is slower and limits how much it can make. That is why cacti grow slowly and why plants with plenty of water do not bother.

Full credit needs the water saving and a cost. The best answers connect the slow growth of desert plants to this trade rather than treating it as a separate fact.

MARKING SHORTCUT

Mark question 2 on the question page and Go Further question 3. The first shows whether they understand the mechanism; the second shows whether they can read a trade-off out of numbers rather than being told about it.

The Instruction Book

NO SCIENCE BACKGROUND NEEDED. NO LAB, NO EQUIPMENT.

Everything you need is on this page and the answer key. Nothing in this lesson is heated, poured, cut or plugged in.

Copy for students: the next three pages. **Keep:** this page and the answer key.

What this lesson does

Students use base pairing to rebuild a missing strand, then follow what happens when a single letter is changed. The lesson deliberately separates DNA from inheritance patterns: this is about what a gene physically is and how a change in it reaches the outside of an organism.

How the period runs

MIN	WHAT YOU DO
0-6	Read the opening lines below. Hand out all three student pages.
6-14	Students read the page and study the ladder.
14-28	Questions 1 to 3. The base pairing task is quick and self-checking.
28-40	Questions 4 to 6.
40-53	Go Further: three mutations, three outcomes.
53-55	Collect all pages.

SAY THIS TO START

Every cell you own contains about six feet of DNA, folded up small enough that you need a microscope to see the container. Written out as letters it would fill a bookshelf. And the whole thing uses an alphabet of exactly four letters. Today you will find out what those letters spell, and what happens when one of them is typed wrong.

The three questions students will ask

THEY ASK	YOU SAY
“What is the difference between DNA and a gene?”	DNA is the whole molecule. A gene is one section of it — the instructions for making one particular protein. Like a chapter in a book.
“Are all mutations bad?”	No. Most do nothing at all. Some are harmful. A few are useful. Which one it is depends entirely on where in the instructions it lands.
“How does DNA decide my eye color?”	It does not, directly. It gives instructions for making proteins, and the proteins do the work. Eye color is the amount and kind of pigment protein made.

FINISHED EARLY?

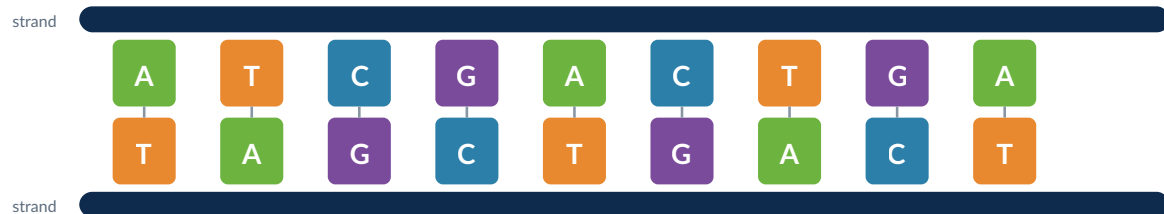
The Go Further page gives three single-letter changes in the same gene and asks students to work out why one is harmless, one is serious and one changes nothing at all.

Read: Four Letters, Twenty Thousand Instructions

BEFORE YOU READ

The ladder below is a short section of DNA. Notice which letters sit opposite which — the pattern is never broken, and question 2 depends on spotting it.

A PAIRS ONLY WITH T · C PAIRS ONLY WITH G



Because the pairing is fixed, either strand alone contains everything needed to rebuild the other.

- 1 DNA is a very long molecule shaped like a twisted ladder. The two sides are the backbones; the rungs are what matter. Each rung is made of two chemicals called **bases** joined in the middle, and there are only four bases in the whole system: A, T, C and G.
- 2 The pairing is rigid and never varies. A always pairs with T. C always pairs with G. No other combination fits. That rule looks like a small chemical detail and it is the reason life can copy itself: if you know one side of the ladder you can rebuild the other exactly, which is precisely what a cell does before it divides.
- 3 The order of those letters along one strand is the information. A **gene** is a section of that sequence — typically a few thousand letters — carrying the instructions for building one particular **protein**. Your DNA holds roughly twenty thousand genes, and a chromosome is one enormously long DNA molecule with many genes on it.
- 4 Proteins do essentially all the work in a body. They form structures like hair, skin and muscle. They act as enzymes that run every chemical reaction you depend on. They carry oxygen, fight infection and send signals. If something is happening inside you, a protein is doing it.
- 5 So the chain runs like this, and it is worth learning as a chain: **DNA → protein → trait**. A gene is read, a protein is built to that specification, and the protein produces the characteristic you can actually observe. Nothing about your DNA reaches the outside world except through a protein.
- 6 A **mutation** is a change in the sequence — a letter swapped, added or deleted. They happen constantly, from copying errors and from damage by things like ultraviolet light. Your cells carry repair machinery that catches most of them, but not all.
- 7 What a mutation does depends entirely on where it lands, and most of them do nothing whatsoever. The change may fall in a stretch that is not part of any gene, or it may change a letter without changing which protein gets built. These are silent, and they are the most common outcome by a wide margin.
- 8 Occasionally a mutation changes the protein. Usually that makes the protein work slightly worse, and sometimes it stops it working at all, which is how many genetic conditions arise. Very rarely, the altered protein does something useful. All three outcomes come from the same kind of event; nothing in the chemistry knows or cares which one it produced.

Questions: The Instruction Book

Answer in the space provided.

- 1 State the two pairing rules for DNA bases.

- 2 One strand is given. Write the matching strand underneath.

Given strand

A T G C C A T G A C

Matching strand

- 3 Explain why a cell only needs one strand of the ladder to rebuild the whole molecule.

- 4 Explain the difference between DNA, a gene and a chromosome.

- 5 A student says “my DNA makes my hair curly.” Rewrite that sentence so it correctly includes the missing step.

- 6 Most mutations have no effect at all. Give two different reasons why that can be true.

Go Further: One Letter, Three Outcomes

EXTEND

THE SITUATION

The same gene in four people. It carries the instructions for one protein. Person 1 is the unchanged version. The others each have a single letter different, shown in bold.

PERSON	SECTION OF THE GENE	PROTEIN PRODUCED	EFFECT ON THE PERSON
1	CAT GGA TCC	Normal, works fully	None — this is the usual version
2	CAT GGG TCC	Identical to person 1	
3	CAT A GA TCC	Slightly different shape, works at about half speed	
4	CAT T GA TCC	Building stops early. No usable protein at all	

- 1 Complete the last column. Predict the effect on each person and explain your reasoning.
- 2 Person 2's DNA is definitely different from person 1's, yet nothing about them is different. Explain how both can be true.
- 3 Person 3 and person 4 have a change in the same position. Explain why one is far more serious than the other.
- 4 All three changes are exactly the same size — one letter. What does that tell you about predicting how serious a mutation will be?

CHALLENGE

Sunlight damages DNA in skin cells constantly, and yet most people do not develop problems from an afternoon outdoors. Using what you know about where mutations land and what cells can do about them, give two separate reasons why. Three or four sentences.

Answer Key: The Instruction Book

QUESTION PAGE

1. A always pairs with T. C always pairs with G.

2. T A C G G T A C T G

Self-checking: any letter opposite an A that is not a T is wrong at a glance. Worth marking first because it takes ten seconds.

3. Because the pairing rule never varies. Each letter on one strand determines exactly one letter opposite it, so one strand contains all the information needed to build the other.

4. DNA is the molecule itself. A gene is one section of that molecule carrying the instructions for one protein. A chromosome is one very long DNA molecule containing many genes.

5. Something like: "My DNA carries instructions for making a protein, and that protein makes my hair curly." The missing step is the protein.

This is the question worth marking. The chain DNA → protein → trait is the single most useful idea in the lesson, and students skip the middle every time.

6. Any two of: the change fell in a stretch of DNA that is not part of any gene; the change did not alter which protein was built; the cell's repair machinery corrected it before it mattered.

GO FURTHER PAGE

1. Person 2: no effect, the protein is identical. Person 3: some effect, the protein works but only at about half speed, so whatever it does happens more slowly. Person 4: serious effect, no usable protein is made at all.

2. Because DNA only affects a person through the protein it produces. Person 2's change did not alter the protein, so there is nothing to pass on to the rest of the body. A difference that never reaches a protein never reaches the person.

Full credit needs the protein as the link. This question and question 5 on the previous page are testing the same idea from opposite directions.

3. Person 3's change alters the protein's shape, so it still works but less well. Person 4's change stops the protein being built at all. A protein working at half speed is a problem; no protein is usually a much bigger one.

4. That the size of a mutation tells you almost nothing about how serious it is. What matters is where it lands and what it does to the protein, not how many letters changed.

This is the best question on the page and the one to mark if you are short on time. It generalizes beyond biology: the size of a change and the size of its effect are different questions.

Challenge. First, most of the damage lands somewhere that does not matter — outside any gene, or in a spot where the change does not alter the protein. Second, cells carry repair machinery that finds and fixes most damage before the cell divides. Problems arise when a mutation lands in an important gene and escapes repair, which is rare per event but becomes likely with enough exposure over enough years.

Full credit needs both reasons. Strong students add that the risk is cumulative, which is exactly why sun protection is framed as a lifetime habit rather than a one-day precaution.

Sorting the Living World

NO SCIENCE BACKGROUND NEEDED. NO LAB, NO EQUIPMENT.

Everything you need is on this page and the answer key. Nothing in this lesson is heated, poured, cut or plugged in.

Copy for students: the next three pages. **Keep:** this page and the answer key.

What this lesson does

Students run six creatures through a dichotomous key, then build a short key of their own. It is one of the few genuinely procedural tasks in science that needs no equipment, it is completely self-checking, and weaker readers do well at it.

How the period runs

MIN	WHAT YOU DO
0-5	Read the opening lines below. Hand out all three student pages.
5-13	Students read the page and the worked key.
13-28	Questions 1 to 3. Question 2 is the main task.
28-40	Questions 4 to 6.
38-48	Go Further: building a key from scratch.
48-50	Collect all pages.

SAY THIS TO START

There are somewhere around nine million kinds of living thing on this planet and we have named maybe a fifth of them. If you find something in a river and want to know what it is, you cannot flip through nine million pictures. So biologists built a tool that gets you there in about six questions, and every one of those questions has only two possible answers. Today you are going to use it, and then build one.

The three questions students will ask

THEY ASK	YOU SAY
"Why not just use a picture guide?"	Because there are far too many species, and many look almost identical. A key asks about features instead, so it works even for something nobody has photographed.
"What if neither answer fits?"	Then you have gone wrong at an earlier step, or the key does not cover your organism. Go back one step and check. That is a normal part of using a key.
"Why is every question a choice of two?"	Because two-way choices cannot be ambiguous, and each one halves what is left. Six questions can separate more than sixty things.

FINISHED EARLY?

The Go Further page gives six shapes with described features and asks students to build a working key for them, which is much harder than following one and shows real understanding.

Read: Nine Million Kinds of Thing

BEFORE YOU READ

Read the passage first, then work through the example key at the bottom of this page. You will use exactly the same method on the next page.

- 1 Biologists sort living things into groups, and the groups sit inside one another like boxes inside boxes. The largest divisions are the **kingdoms** — animals, plants, fungi and several groups of single-celled organisms. Each kingdom divides again and again, ending at the **species**, which is the individual kind.
- 2 Every species gets a two-word scientific name, written in italics, made of its group name and its own name. A red fox is *Vulpes vulpes* in every country on Earth. That matters more than it sounds: the same animal is called a cougar, a puma, a mountain lion and a panther in different parts of one continent, and all four are *Puma concolor*.
- 3 The grouping is not arbitrary. Things are placed together because they share features, and shared features usually mean shared ancestry. So the classification is really a family tree, which is why it has held up as new evidence arrived rather than needing to be thrown out.
- 4 Identifying something is a separate problem from classifying it. For that, biologists use a **dichotomous key**. *Dichotomous* means divided in two, and that is the entire principle: every step offers exactly two options, and each one either names your organism or sends you to another numbered step.
- 5 Two-way choices are used because they cannot be ambiguous. There is no room for “sort of”. And each choice removes half of what is left, so the number of steps needed grows very slowly even as the number of species grows quickly — six steps can separate more than sixty organisms.
- 6 A good key only ever asks about features you can actually see on the specimen in front of you. Counting legs, checking for wings, looking at whether leaves are in pairs. A key that asks how long the animal lives, or what it eats, is useless in the field even if the information is correct.

WORKED EXAMPLE — DONE FOR YOU

Follow this key for a **spider**. Start at step 1 every time.

- | | |
|----|--|
| 1a | Has more than six legs → go to 2 |
| 1b | Has six legs or fewer → go to 3 |
| 2a | Has exactly eight legs → Spider |
| 2b | Has many more than eight legs → Centipede |
| 3a | Has wings → Beetle |

A spider has eight legs, so step 1 sends you to 2, and step 2a names it. Two questions, done.

Questions: Sorting the Living World

Use the key below for questions 1 and 2. Always start at step 1.

KEY TO SIX SMALL ANIMALS

1a	Has a hard outer shell → go to 2
1b	No hard outer shell → go to 3
2a	Shell in a single coil → Snail
2b	Shell in two hinged halves → Mussel
3a	Has jointed legs → go to 4
3b	No legs at all → Earthworm
4a	Six legs → go to 5
4b	More than six legs → Woodlouse
5a	Wings present → Housefly
5b	No wings → Ant

1 Identify each animal using the key. Write the steps you used, like this: 1b, 3a, 4a, 5b.

THE ANIMAL IN FRONT OF YOU	STEPS YOU USED	WHAT IT IS
No shell, jointed legs, six legs, has wings		
Hard shell in two hinged halves		
No shell, no legs at all		
No shell, jointed legs, ten legs		

2 A student starts at step 3 instead of step 1 and identifies a snail as an earthworm. Explain what went wrong.

3 Every step in the key offers exactly two choices. Explain why that is better than offering four.

4 Why is “has jointed legs” a better question for a key than “lives in the soil”?

5 A cougar, a puma and a mountain lion are all the same animal. Explain why scientists give it the single name *Puma concolor*.

Go Further: Build Your Own Key

THE TASK

Six imaginary creatures have been collected. Your job is to write a key that separates all six. This is harder than following one, and there is more than one correct answer.

CREATURE	LEGS	WINGS	BODY COVERING	TAIL
Vorm	0	No	Smooth skin	No
Glib	4	No	Scales	Yes
Trask	4	No	Fur	Yes
Pell	4	No	Fur	No
Skirl	6	Yes	Smooth skin	No
Nunk	6	No	Smooth skin	No

1 Write your key. Use as many steps as you need. Every creature must have exactly one route to it.

STEP	IF YES...	IF NO...
1		
2		
3		
4		
5		

2 Test your key on the **Pell**. Write down the steps it takes. Does it arrive at the right creature?

3 Which single feature was most useful as your first question? Explain why.

CHALLENGE

A seventh creature is found: **4 legs, no wings, fur, no tail**. Run it through your key. It should come out as a Pell, but it is a different species. Explain why this is not a fault in your key, and say what you would need in order to fix it. Three or four sentences.

Answer Key: Sorting the Living World

QUESTION PAGE

DESCRIPTION	STEPS	ANSWER
No shell, jointed legs, six legs, wings	1b, 3a, 4a, 5a	Housefly
Hard shell, two hinged halves	1a, 2b	Mussel
No shell, no legs	1b, 3b	Earthworm
No shell, jointed legs, ten legs	1b, 3a, 4b	Woodlouse

2. They skipped the first question, which is the one that separates shelled animals from unshelled ones. A key only works if you start at step 1 and follow it in order, because every later step assumes the earlier ones have already been answered.
- This is the most useful question on the page. Students who understand it stop making the mistake immediately.*
3. Because a two-way choice cannot be ambiguous — there is no room for a “sort of” answer. With four options it is much easier for two of them to overlap or for none to fit exactly.
4. Because you can see jointed legs on the specimen in front of you. Where something lives is not visible on the animal itself, so you would have to already know what it was to answer.
5. Because common names differ from place to place and the same name is used for different animals elsewhere. One scientific name means every biologist in the world is certainly talking about the same species.

60 FURTHER PAGE

ONE CORRECT KEY – OTHERS ARE EQUALLY VALID

1	Has legs? No → Vorm Yes → go to 2
2	Six legs? Yes → go to 3 No (four) → go to 4
3	Has wings? Yes → Skirl No → Nunk
4	Covered in fur? No (scales) → Glib Yes → go to 5
5	Has a tail? Yes → Trask No → Pell

Mark the student's own key, not this one. The three things to check: every creature is reachable, no creature can be reached by two different routes, and every question can be answered from the table.

2. Using the key above: 1 yes → 2 no → 4 yes → 5 no → Pell. Any route that arrives at Pell and only at Pell is correct.
3. Number of legs is the strongest opening question, because it splits the six into groups of 1, 3 and 2 straight away. Any answer that explains the split in terms of how evenly it divides the group is full credit.
- Students who chose “has fur” first end up with a lopsided key that still works. It is worth pointing out that a working key and an efficient key are different things.*
- Challenge.** It is not a fault, because a key can only separate the creatures it was built from. The new creature matches the Pell on every feature in the table, so no key built from that table could possibly tell them apart. To fix it you would need to find a feature that differs between them — size, color, number of eyes, anything observable — and add a step using it.
- Full credit needs both halves: the key is not wrong, and the fix is a new distinguishing feature rather than a rearrangement of the existing steps.*

Boom and Crash

NO SCIENCE BACKGROUND NEEDED. NO LAB, NO EQUIPMENT.

Everything you need is on this page and the answer key. Nothing in this lesson is heated, poured, cut or plugged in.

Copy for students: the next three pages. **Keep:** this page and the answer key.

What this lesson does

Students read two population curves that rise and fall out of step with each other, then work out why the predator peak always comes after the prey peak. The lesson is about numbers over time rather than about who eats whom, and it ends on what happens when nothing limits a population at all.

How the period runs

MIN	WHAT YOU DO
0-6	Read the opening lines below. Hand out all three student pages.
6-14	Students read the page and study the graph.
14-28	Questions 1 to 4, all read off the graph.
28-40	Questions 5 and 6.
40-53	Go Further: the deer on the plateau.
53-55	Collect all pages.

SAY THIS TO START

Here is a puzzle. On an island, the rabbit population goes up and down in a regular cycle, and so does the fox population. But the fox peak always arrives about two years after the rabbit peak. Never before. Never at the same time. Always behind. Today you will work out why the delay is there, and why it could not possibly happen the other way round.

The three questions students will ask

THEY ASK	YOU SAY
"Why does the predator peak come later?"	Because predators can only increase after there is plenty to eat. The food has to arrive first. That gap is the whole lesson.
"What is carrying capacity?"	The largest population an area can support long term. Above it there is not enough food, water or space, so numbers fall back.
"Do predators wipe out their prey?"	Almost never. As prey get scarce the predators starve and decline, which lets the prey recover. The system corrects itself.

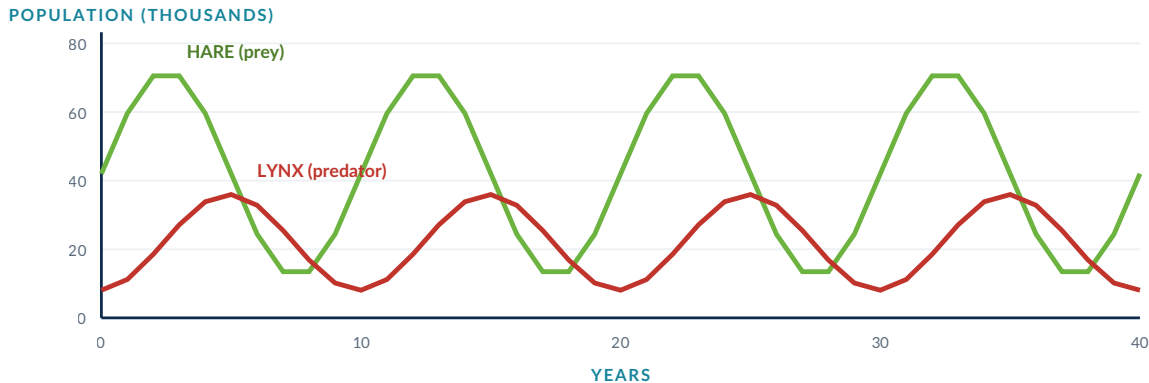
FINISHED EARLY?

The Go Further page tells the story of a deer herd whose predators were removed. Students read the numbers and work out why the outcome was the opposite of what was intended.

Read: Why the Predator Peak Comes Late

BEFORE YOU READ

Two populations on the same island over forty years. Look at where each line peaks before you read anything. Questions 1 to 4 all come straight off this graph.



1 No population grows forever. Given unlimited food and space, numbers would rise faster and faster without end, and that never happens anywhere. Something always stops it, and the things that do the stopping are called **limiting factors**: food, water, space, disease, and being eaten.

2 The largest population an area can support indefinitely is its **carrying capacity**. It is not a fixed number carved into the landscape — a wet year raises it and a drought lowers it — but at any moment there is a ceiling, and populations that go above it come back down.

3 The graph shows a classic pattern. Hares rise, lynx rise, hares fall, lynx fall, and the whole cycle repeats about every ten years. Neither line is steady and neither is random. This is one of the most reliably repeated patterns in ecology and it has been recorded from fur trading records going back nearly two centuries.

4 The order of events is the part worth slowing down for. When hares are plentiful, lynx have abundant food, so more of their young survive and the lynx population climbs. But that climb takes time — animals have to be born and grow — so the lynx peak arrives about two years *after* the hare peak. The food must come first. It could not be otherwise.

5 Then the pressure reverses. Large numbers of lynx eat hares faster than hares can reproduce, so the hare population falls. Now there are many lynx and little food, so lynx begin to starve and their numbers drop too. With few lynx left, the surviving hares breed freely and rise again, and the whole cycle restarts.

6 Notice that nothing is being managed and nobody is deciding anything. The lag between the two populations is what produces the cycle, and it is why predators almost never eliminate their prey: as prey become scarce, the predators decline first, which takes the pressure off before the last prey animal is gone.

7 It is worth separating two kinds of limit. **Density-dependent** factors bite harder when a population is crowded — disease spreads faster, food runs short sooner, predators concentrate where prey are thick. **Density-independent** factors, like a flood, a fire or a hard freeze, hit the same regardless of how many there are.

8 And when a limiting factor is taken away, the result is rarely what people expect. A population released from its ceiling does not settle at a comfortable new level. It overshoots, strips its food supply faster than the supply can regrow, and then crashes to something lower than where it started — because the habitat itself has been damaged on the way through.

Questions: Boom and Crash

Questions 1 to 4 refer to the graph on the reading page.

1 Roughly how many years pass between one hare peak and the next?

2 Which population reaches a higher number, and roughly what are the two peak values?

3 Does the lynx peak come before or after the hare peak? By roughly how many years?

4 Explain why the lynx peak **has** to come after the hare peak and could never come before it.

This is the most important question on the page.

5 Explain what happens to the hare population when lynx numbers are high, and then what that does to the lynx.

6 Classify each limiting factor. Write **DD** for density-dependent or **DI** for density-independent.

LIMITING FACTOR	DD OR DI	LIMITING FACTOR	DD OR DI
A disease spreading through the herd		A forest fire	
Competition for food		An unusually cold winter	

Go Further: The Deer on the Plateau

THE SITUATION

In 1907 a deer herd lived on a large forested plateau alongside wolves and mountain lions. To increase deer numbers for hunting, the predators were deliberately removed over the following years. These are the recorded deer numbers.

YEAR	1907	1915	1920	1924	1926	1931	1939
Deer	4,000	25,000	60,000	100,000	40,000	20,000	10,000

- 1 Describe what happened to the deer between 1907 and 1924.
-
-
-
- 2 Describe what happened between 1924 and 1939. How does the 1939 figure compare with 1907?
-
-
-
- 3 Removing the predators was supposed to give more deer. Explain why it produced fewer.
-
-
-
-
- 4 By 1924 the plateau could no longer feed 100,000 deer. Suggest what condition the forest itself was in by then, and why that made the crash worse.
-
-
-
-

CHALLENGE

The final population of 10,000 was well above the 4,000 the plateau started with, and yet ecologists describe the whole episode as a failure. Give one argument that 10,000 deer counts as a success, and one argument that it does not. Then say which you find more convincing and why. Four or five sentences.

Answer Key: Boom and Crash

QUESTION PAGE

1. About ten years.
2. The hares. Hares peak at roughly 70,000 and lynx at roughly 36,000.
Accept anything close. Reading approximate values off a curve is the skill; precision is not.
3. After, by about two years.
4. Because lynx numbers can only rise once there is plenty of food, and hares are that food. The hares have to become abundant first, and then lynx need time to breed and for their young to survive. A cause cannot come after its effect, so the peak in food must come before the peak in the animals that live on it.
This is the question that carries the lesson. Full credit needs the idea that the food has to arrive first and that breeding takes time. Students who say only "lynx eat hares" have described the relationship without explaining the delay.
5. High lynx numbers eat hares faster than hares can reproduce, so the hare population falls. That leaves many lynx with little food, so lynx begin to starve and their numbers fall too, which eventually lets the hares recover.
6. DD: disease spreading through the herd, competition for food. DI: forest fire, unusually cold winter.
The test is whether being crowded makes it worse. A fire burns the same whether there are ten deer or ten thousand.

60 FURTHER PAGE

1. It rose enormously, from 4,000 to 100,000 in seventeen years — a twenty-five-fold increase.
 2. It crashed, falling from 100,000 to 40,000 in two years and continuing down to 10,000 by 1939. That is still higher than the 4,000 in 1907, but only a tenth of the peak.
 3. Removing the predators removed a limiting factor, so the deer population overshot what the plateau could feed. They ate the vegetation faster than it could regrow, then starved in large numbers. The predators had been holding the population near a level the habitat could actually sustain.
Full credit needs the overshoot. "They ran out of food" is only half the answer — the question is why they were allowed to get to a number where that was inevitable.
 4. Heavily overgrazed — young trees eaten, ground cover stripped, little left within reach. That made the crash worse because the carrying capacity itself had been lowered, so the land could now support fewer deer than before the whole episode began.
This is the best question on the page. It requires seeing that the ceiling itself moved, not just that the population hit it.
- Challenge.** For: 10,000 is still more than double the starting 4,000, so by the original aim of having more deer, it worked. Against: tens of thousands of deer starved, the forest was seriously damaged, the land now supports fewer deer than it could have, and the result was completely unintended. Either position can earn full marks if it is argued from the numbers.
Look for an actual argument on both sides before the student picks one. A student who only argues the side they believe has answered half the question, however well they argue it.

How Far Is Far

NO SCIENCE BACKGROUND NEEDED. NO LAB, NO EQUIPMENT.

Everything you need is on this page and the answer key. Nothing in this lesson is heated, poured, cut or plugged in.

Copy for students: the next three pages. **Keep:** this page and the answer key.

What this lesson does

Students work with a unit that is a distance and a time at once, then use it to reach the conclusion that every telescope is a machine for looking into the past. The scale calculations are simple division and the payoff is one of the genuinely startling ideas in science.

How the period runs

MIN	WHAT YOU DO
0-6	Read the opening lines below. Hand out all three student pages.
6-14	Students read the page.
14-28	Questions 1 to 4.
28-40	Questions 5 and 6.
40-53	Go Further: the travel times table.
53-55	Collect all pages.

SAY THIS TO START

Look at the Sun — not directly, obviously. The light hitting your face left it eight minutes ago. If the Sun vanished right now, you would carry on seeing it for another eight minutes, quite normally, before anything changed. That is not a trick. Every single thing you have ever looked at, you have seen late. Today you will find out how late, and for some things the answer is millions of years.

The three questions students will ask

THEY ASK	YOU SAY
“Is a light-year a time or a distance?”	A distance — how far light travels in one year, about six trillion miles. The word year in it confuses everybody, so say this early.
“Are we really seeing the past?”	Yes, always, for everything. Light takes time to arrive. For this room it is a few billionths of a second; for distant galaxies it is millions of years.
“Could a star we can see already be gone?”	Yes. If a star 500 light-years away exploded 200 years ago, nobody on Earth will know for another 300 years.

FINISHED EARLY?

The Go Further page gives distances to six objects and asks students to convert them into how far back in time they are looking.

Read: Everything You See Is Late

- 1 Distances in space break ordinary units. Writing the distance to the nearest star in miles means writing twenty-five trillion, and the number is so large it stops carrying any meaning. So astronomers measure with the fastest thing there is.
- 2 Light travels 186,000 miles every second. In one year it covers about six trillion miles, and that distance is called a **light-year**. It is worth being clear immediately: a light-year is a *distance*, not a length of time. The word year is in there because of how the distance is defined, and it misleads almost everybody the first time.
- 3 Using it, the numbers become manageable. The Moon is 1.3 light-seconds away. The Sun is 8.3 light-minutes. Neptune is about 4 light-hours. The nearest star beyond our own, Proxima Centauri, is 4.2 light-years. The center of our galaxy is 26,000 light-years away, and the nearest large galaxy, Andromeda, is 2.5 million.
- 4 Now the consequence, which follows from the definition and is still hard to accept. Light takes time to travel, so what you see is not the object now. It is the object as it was when the light left it. Looking at the Sun shows you the Sun of eight minutes ago. Looking at Proxima Centauri shows you a star as it was when you were in elementary school.
- 5 Push that further. Andromeda's light left it 2.5 million years ago, before any human being existed. When you look at that smudge in the night sky, you are looking at an object as it was before your species did. Nobody alive knows what Andromeda looks like today and nobody ever will.
- 6 This turns every telescope into a time machine pointed backward, and astronomers use it deliberately. To study what the universe was like long ago, you do not need a theory about the past — you point at something far enough away that its light has taken that long to arrive, and simply look at it.
- 7 The same logic applies to everything, just at scales too small to notice. The light from this page reaches your eye in about a billionth of a second, so you are reading a page as it was a billionth of a second ago. Nothing is ever seen in the present. Space is just large enough to make the delay obvious.
- 8 And the sky is not a flat ceiling. Two stars sitting side by side in the same constellation can be wildly different distances away, their light arriving from wildly different centuries, happening to line up from where we stand. A constellation is a coincidence of viewing angle, not a group of neighbors.

Questions: How Far Is Far

Answer in the space provided.

- 1 Is a light-year a measure of time or of distance? Explain what it actually measures.

- 2 If the Sun stopped shining this instant, how long would it be before anyone on Earth noticed? Explain.

- 3 Proxima Centauri is 4.2 light-years away. Explain what you are actually seeing when you look at it.

- 4 Explain why astronomers describe a powerful telescope as a way of looking into the past.

- 5 A star 500 light-years away exploded 200 years ago. Would we know about it yet? Explain your reasoning.

- 6 Two stars appear side by side in the same constellation. Explain why they may not be anywhere near each other.

Go Further: Looking Backward

EXTEND

THE DATA

Distances to six objects, and how long light from each takes to reach us. Complete the table.

OBJECT	DISTANCE	LIGHT TAKES	SO WE SEE IT AS IT WAS...
The Moon	239,000 miles	1.3 seconds	
The Sun	93 million miles	8.3 minutes	
Neptune	2.8 billion miles	4.1 hours	
Proxima Centauri	4.2 light-years		
Center of our galaxy	26,000 light-years		
Andromeda galaxy	2.5 million light-years		

- 1

Complete the last two columns. For the final three rows the distance tells you the answer directly – explain why in one sentence.
- 2

The center of our galaxy is 26,000 light-years away. Humans were living in the Ice Age when that light set out. Write one sentence explaining what that means about the photograph an astronomer takes of it today.
- 3

A spacecraft travels at 40,000 mph, the fastest we have ever managed. Light covers a light-year in one year. Roughly how many years would that spacecraft take to cross one light-year? Use 6 trillion miles and show your working.

CHALLENGE

An astronomer says: “We will never know what Andromeda looks like today.” That sounds like a limit of our telescopes, and it is not. Explain why no telescope ever built, however powerful, could solve this. Three or four sentences.

Answer Key: How Far Is Far

QUESTION PAGE

1. Distance. It is how far light travels in one year, about six trillion miles.

If a student writes "time", that is the expected error and worth correcting to the whole class. The word year does the damage.

2. About 8.3 minutes. Light already on its way would keep arriving normally until the last of it reached us.

3. The star as it was 4.2 years ago, because that is how long its light took to arrive. Nothing about it in the last 4.2 years has reached us yet.

4. Because light takes time to travel, so a more distant object is seen as it was longer ago. Pointing a telescope at something far away is a way of observing what the universe was actually like at that time, rather than reasoning about it.

5. No. The light needs 500 years to reach us and the explosion was only 200 years ago, so it is still 300 years away. Nobody will see it until then.

The arithmetic is the easy part. The idea that an event can have already happened and not yet be knowable is the difficult part, and it is worth a minute of discussion.

6. Because a constellation is only a pattern seen from where we happen to stand. Two stars can line up from this angle while being enormously different distances away, with one far behind the other.

GO FURTHER PAGE

OBJECT	LIGHT TAKES	WE SEE IT AS IT WAS
Moon	1.3 seconds	1.3 seconds ago
Sun	8.3 minutes	8.3 minutes ago
Neptune	4.1 hours	4.1 hours ago
Proxima Centauri	4.2 years	4.2 years ago
Center of our galaxy	26,000 years	26,000 years ago
Andromeda	2.5 million years	2.5 million years ago

1. Because a light-year is defined as the distance light travels in one year, so a distance given in light-years is already the travel time in years.

2. The photograph shows the galactic center as it was 26,000 years ago, during the Ice Age. It is a picture of the deep past, not of today.

3. $6,000,000,000,000 \div 40,000 = 150,000,000$ hours. Divided by 24 that is 6.25 million days, and divided by 365 that is about **17,000 years** to cross a single light-year.

Accept anything in the range 16,000 to 18,000 with working shown. The number is the point: our fastest spacecraft would need 17,000 years to cover a distance light crosses in twelve months.

Challenge. Because the problem is not how much light a telescope can collect — it is that the light carrying today's Andromeda has not arrived and will not arrive for 2.5 million years. A telescope can only work with light that has already reached us. No amount of magnification or sensitivity creates information that is not here yet.

This is the payoff of the lesson. Full credit needs the recognition that this is a limit of information arriving, not of instrument quality.

MARKING SHORTCUT

Mark question 5 on the question page and the Challenge. Both test whether the student really believes the central claim or has only learned to repeat it. The table is self-checking.

Where the Power Comes From

NO SCIENCE BACKGROUND NEEDED. NO LAB, NO EQUIPMENT.

Everything you need is on this page and the answer key. Nothing in this lesson is heated, poured, cut or plugged in.

Copy for students: the next three pages. **Keep:** this page and the answer key.

What this lesson does

Students sort resources by whether they run out, then compare five ways of generating electricity across four different measures. There is no clean winner in the table, which is deliberate: the lesson is about reasoning with trade-offs rather than arriving at a favorite.

How the period runs

MIN	WHAT YOU DO
0-5	Read the opening lines below. Hand out all three student pages.
5-13	Students read the page.
13-26	Questions 1 to 3. The sorting table is quick.
26-38	Questions 4 to 6.
38-48	Go Further: the comparison table.
48-50	Collect all pages.

SAY THIS TO START

Everything you used today came out of the ground or off the land. The metal in the bus, the concrete in this building, the fuel that got you here, the electricity in these lights. Some of those we can keep taking indefinitely. Some of them exist in a fixed amount and every ton used is a ton that is never coming back. Today you will work out which is which, and why that distinction is harder to apply than it looks.

The three questions students will ask

THEY ASK	YOU SAY
“What makes something renewable?”	It is replaced as fast as we use it, or faster. Sunlight and wind arrive whether we use them or not. Coal took three hundred million years to form.
“Is wood renewable?”	It depends entirely on how fast it is cut. Replanted at the rate it is felled, yes. Cleared faster than it regrows, no. The rate decides, not the material.
“Which one is best?”	There is no single answer, and the table on the Go Further page is designed to show that. Every option is best at something and worst at something else.

FINISHED EARLY?

The Go Further page compares five sources across cost, reliability, land use and emissions. No column has the same winner, which is the point.

Read: What Runs Out and What Does Not

- 1 A **natural resource** is anything we take from the Earth and use: water, timber, iron, sand, oil, sunlight. The most useful division is by whether taking it makes less of it available for the future.
- 2 A **renewable** resource is replaced at least as fast as we use it. Sunlight arrives every morning whether we capture it or not. Wind keeps blowing. Rivers keep flowing. Using these does not reduce tomorrow's supply in any meaningful sense.
- 3 A **non-renewable** resource exists in a fixed amount and is not being made again on any timescale that matters to us. Coal formed from plants buried around three hundred million years ago under conditions that no longer occur. Oil and natural gas formed from marine organisms over similar spans. Every barrel used is one that will not be replaced.
- 4 Metals sit in an awkward middle. There is a fixed amount of copper in the Earth's crust, so ore is non-renewable. But copper does not get destroyed when it is used — it ends up in wire, pipe and scrap, and can be melted down and used again indefinitely. Recycling does not create more copper; it keeps the copper we already dug up in circulation instead of in landfill.
- 5 Some resources move between the two categories depending on how they are treated, and timber is the clearest case. A forest replanted at the rate it is cut is genuinely renewable and can run forever. The same forest cleared faster than it regrows is being mined, not harvested. What decides is the rate, not the material.
- 6 Fresh water is the same story and the more urgent one. Rain replenishes rivers and lakes continuously, so surface water is renewable. But some underground reserves were filled during the last ice age and refill at a few inches a year, while being pumped out far faster. That water is being treated as renewable and behaving as though it is not.
- 7 Turning any of this into electricity involves a choice with no clean answer. Fossil fuels are cheap to build for, work whenever you want them and release carbon dioxide. Solar and wind release almost nothing while running but only work when the sun shines or the wind blows. Nuclear runs constantly with very low emissions and produces waste that stays dangerous for a very long time.
- 8 So the honest position is that every option is best on some measure and worst on another, and the right mix depends on where you are — how much sun, how much wind, what rivers you have, what you can afford, and what you are prepared to live with. Anyone who tells you one source is simply the best answer has stopped looking at the table.

Questions: Where the Power Comes From

Answer in the space provided.

1 Sort each resource. Write **R** for renewable or **N** for non-renewable.

RESOURCE	R OR N	RESOURCE	R OR N	RESOURCE	R OR N
Coal		Wind		Natural gas	
Sunlight		Iron ore		Flowing rivers	
Crude oil		Copper ore		Geothermal heat	

2 Explain why coal is described as non-renewable when it is technically still forming somewhere right now.

3 Timber can be renewable or non-renewable depending on how it is used. Explain both cases.

4 Recycling copper does not create any new copper. Explain why it still matters.

5 Some underground water refills at a few inches a year but is pumped much faster. Explain why that is a problem even though rain has not stopped.

6 Give one advantage and one disadvantage of solar power.

Go Further: No Column Has the Same Winner

EXTEND

THE DATA

Five ways of generating the same amount of electricity, compared on four measures. Lower cost is better, higher reliability is better, less land is better, lower emissions are better.

SOURCE	COST	RELIABILITY	LAND NEEDED	CARBON DIOXIDE
Coal	Low	Very high	Small	Very high
Natural gas	Low	Very high	Small	High
Solar	Medium	Low — daytime only	Very large	Very low
Wind	Medium	Low — windy days only	Large	Very low
Nuclear	High to build	Very high	Small	Very low

- 1 Which source wins on carbon dioxide? Which wins on reliability? Are they the same source?
- 2 No source wins on all four measures. Explain what that means for anyone trying to choose one.
- 3 A country with lots of sunshine, plenty of empty land and very little money to spend up front is choosing. Which would you recommend, and which measure did you decide to sacrifice?
- 4 A country with little sun, little wind, no spare land and a strict emissions target is choosing. Which would you recommend, and what is the cost of that choice?

CHALLENGE

Solar and wind both score “low” on reliability because they only work at certain times. Suggest one way a country could use a lot of solar and wind without the lights going out at night, and explain what that solution would add to the table — because it is not free. Three or four sentences.

Answer Key: Where the Power Comes From

QUESTION PAGE

1. Renewable: sunlight, wind, flowing rivers, geothermal heat. Non-renewable: coal, crude oil, natural gas, iron ore, copper ore.
2. Because it forms over roughly three hundred million years, which is unimaginably slower than we use it. A resource counts as renewable only if it is replaced as fast as it is consumed, and coal is not remotely close.
3. Renewable if the forest is replanted at the same rate it is cut, so the supply lasts indefinitely. Non-renewable if it is cleared faster than it can regrow, because then the total amount is falling and will eventually be gone. The rate decides, not the material.
This is the best question on the page. It shows the categories are about how a resource is managed, not just what it is made of.
4. Because it keeps copper that has already been mined in use instead of in landfill, so less new ore has to be dug up. It cannot increase the total copper on Earth, but it can make the fixed supply last far longer.
5. Because that water is being taken out much faster than it is replaced, so the reserve is shrinking regardless of rainfall. It behaves like a non-renewable resource even though water in general is renewable — the rate is what matters.
6. Advantage: very low carbon dioxide, and sunlight never runs out. Disadvantage: only works in daylight, and needs a very large area of land.

60 FURTHER PAGE

1. Solar, wind and nuclear all tie on carbon dioxide at very low. Coal, gas and nuclear all tie on reliability at very high. Nuclear is the only source that appears in both lists.
Students who spot that nuclear wins both are reading the table properly. The obvious follow-up — so why not build only nuclear? — is answered by the cost column.
 2. That there is no single right answer and any choice means accepting a weakness. Choosing a source is deciding which disadvantage you are most willing to live with.
 3. Solar. Plenty of sun makes the reliability problem smaller, and plenty of empty land makes the land problem smaller. What is sacrificed is reliability at night, which would have to be managed some other way.
Accept wind if the reasoning fits the conditions given. The mark is for naming the sacrifice, not for the choice.
 4. Nuclear. It is the only option that is very high on reliability and very low on emissions while needing little land. The cost is the high price of building it, and the long-lived waste it produces.
- Challenge.** Store the electricity when there is a surplus — in large batteries, or by pumping water uphill into a reservoir and letting it back down through turbines at night. That would add a new row of costs to the table: the storage itself is expensive, takes materials and land to build, and some energy is always lost in storing and recovering it.
Accept batteries, pumped storage, or connecting to a wider grid so somewhere else supplies the shortfall. Full credit needs the recognition that the solution has its own costs rather than removing the problem.

Emergency Day Checklist

PRINT THIS PAGE AND LEAVE IT ON TOP

Fill in the blanks once, at the start of the year, and this page works for every absence after that.

Note for the substitute

My name

Room / period

Teach this lesson

Lesson #_____ or any lesson from the index

Teacher next door

Lab / storeroom policy

Keep it closed. Nothing in this pack needs equipment.

Students who leave early

Accommodations

Seating, extra time, reader

Fire drill meeting point

Leave the work

Anything else

SUBSTITUTE: BEFORE YOU LEAVE, DO THESE THREE THINGS

1. Stack the collected work on the teacher's desk with this page on top.
2. Write the names of two or three students who were especially helpful. Teachers genuinely act on this.
3. Note anything the teacher should know, in one line. Not a report. One line.

Substitute's note back to the teacher

Lesson taught

Got through

All of it / Most of it / Page 1 only

Helpful students

Anything to know

REUSE NOTE

This pack is designed to be printed once and reused across many absences. Store the briefing pages and answer keys separately from the student pages so the wrong sheets never get photocopied.

Standards Alignment

NGSS MIDDLE SCHOOL · GRADES 6-8

Each lesson is mapped to the closest matching middle school performance expectation. Most lessons also touch others; the primary alignment is listed for clarity.

#	LESSON	FOCUS	NGSS
31	Rub, Stick, Spark	Static electricity, charge and attraction	MS-PS2-5
32	Faster, Slower, Stop	Reaction rate, catalysts and energy changes	MS-PS1-6
33	Trading Distance for Force	Levers, ramps, pulleys and mechanical advantage	MS-ETS1-3
34	Ones and Zeros	Digital and analog signals, noise and copying	MS-PS4-3
35	Water Going Up	Plant transport, stomata and transpiration	MS-LS1-3
36	The Instruction Book	DNA, genes, proteins and mutations	MS-LS3-1
37	Sorting the Living World	Classification and dichotomous keys	MS-LS4-2
38	Boom and Crash	Populations, limiting factors and carrying capacity	MS-LS2-1
39	How Far Is Far	Stars, light-years and the scale of the universe	MS-ESS1-3
40	Where the Power Comes From	Natural resources, renewable and non-renewable	MS-ESS3-1

Coverage by domain

DOMAIN	LESSONS	WHAT IS COVERED
Physical Science	31, 32, 33, 34	Static electricity and charge, reaction rate and energy changes, levers and mechanical advantage, digital and analog signals
Life Science	35, 36, 37, 38	Plant transport and transpiration, DNA and mutations, classification and dichotomous keys, populations and carrying capacity
Earth and Space Science	39, 40	Light-years and the scale of the universe; renewable and non-renewable resources

HOW THIS PACK RELATES TO THE REST OF THE SERIES

Volumes 1, 2 and 3 cover thirty different lessons between them. Nothing in Volume 4 repeats any of them. Together the four volumes give forty emergency science lessons with no overlap at all — enough to cover an absence every fortnight for two full school years without a class ever seeing the same lesson twice.

This resource is independently created and is not affiliated with, endorsed by, or reviewed by any state agency or standards body. NGSS codes are provided to help teachers document the day's instruction.

Terms of Use and Credits

THANK YOU

Thank you for buying Volume 4. It was written by a practicing teacher with more than twenty years in secondary classrooms, and every lesson is built the way I would want a sub plan built if I were the one waking up sick.

If it saved you a morning, a rating on the listing genuinely helps other teachers find it, and earns you credit toward your next purchase.

Getting the most out of this pack

Print it before you need it.	The whole point is that it works on a morning when you cannot get out of bed to print anything.
Store it in two piles.	Student pages in one, briefing pages and answer keys in the other, so the wrong sheets never get copied by mistake.
Tell your office where it is.	On a genuine emergency day, the person who needs to find this is not you.
Use them when you are present too.	These work as standalone skill lessons, review days, and non-lab days when the lab is booked out.

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OTHER PACKS IN THIS SERIES

- Emergency Sub Plans · Middle School Science, **Volume 1** · ten different lessons, no lab required, 57 pages
- Emergency Sub Plans · Middle School Science, **Volume 2** · ten more lessons, no lab required, 57 pages
- Emergency Sub Plans · Middle School Science, **Volume 3** · ten more lessons, no lab required, 57 pages
- Emergency Sub Plans · Middle School Math · ten no-prep lessons, 57 pages
- Emergency Sub Plans · Middle School ELA · ten no-prep lessons, no novel required, 57 pages
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